

ETCD Backup & Restore in Kubernetes

What is etcd?

- etcd is a distributed key-value store that holds all cluster data.
- 👉 etcd = Brain of Kubernetes (stores entire cluster state)

What Does etcd Store?

- 📦 Pods, Deployments, Services
- 🔑 Secrets & ConfigMaps
- 🔑 RBAC policies
- 🧩 CRDs (Custom Resources)
- 🌐 Cluster state & events
- 👉 It is the single source of truth for Kubernetes ⚡

Why etcd Backup is Important?

- If etcd fails ❌ → Kubernetes control plane stops working

Risks:

- Accidental deletion
- Control plane failure
- Disaster recovery
- 👉 Without etcd → Cluster unusable 🚨

Why Backup etcd?

- Recover cluster quickly
- Avoid data loss
- Ensure high availability

Backup Strategy Overview

etcd Snapshot 📦 → Store **Backup** 💾 → **Restore When Needed** 🔁

🧠 Backup Strategy

🕒 Backup Frequency

- ✅ Daily backups
- ✅ Before upgrades
- ✅ Before major changes

⚙️ Automation

- CronJobs
- Scripts
- Backup tools

🔧 etcdctl (CLI Tool)

- 👉 Used for:
 - Backup (snapshot)
 - Restore data
 - Cluster Health check
 - Get/Set values

🔥 Take etcd Backup (Snapshot)

```
ETCDCTL_API=3 etcdctl snapshot save /backup/etcd-snapshot.db \  
--endpoints=https://127.0.0.1:2379 \  
--cacert=/etc/kubernetes/pki/etcd/ca.crt \  
--cert=/etc/kubernetes/pki/etcd/server.crt \  
--key=/etc/kubernetes/pki/etcd/server.key
```

📦 What This Does

- Saves full etcd database, Stored as `.db` file

📦 What Snapshot Contains

generated by "Markdown to PDF Fast Converter" 🌟 <https://dub.sh/79wTuSy>

- Full cluster state
- All Kubernetes objects
- Configuration data

Restore etcd from Snapshot

```
ETCDCTL_API=3 etcdctl snapshot restore /backup/etcd-snapshot.db \
  --data-dir /var/lib/etcd-restore
```

Restore Flow

Snapshot  → Restore Command  → New etcd Data Dir → Cluster Recovery 

etcd in EKS (Managed Kubernetes)

-  Important :  You CANNOT access etcd directly in EKS
-  Why?
 - AWS manages etcd internally
 - Automatic backups handled by AWS
 - Hidden for security & stability

What You Should Do:

- Use Velero
- Use EBS snapshots

What to Say in Interview

- In EKS, etcd is managed by AWS . We use Velero and snapshots for workload backups

Velero (Backup tool for Kubernetes apps & volumes)

- Velero is an open-source tool to:

- backup Kubernetes resources
- Persistent volumes
- Restore workloads
- Migrate clusters

Use Cases

 Scenario	 What You Do	 Why It Matters	 Benefit
 Cluster Upgrade	 Take etcd snapshot before upgrade	 Protect cluster state before risky changes	 Safe rollback if upgrade fails
 Disaster Recovery	 Restore from etcd backup	 Recover entire cluster state (Pods, Secrets, configs)	 Fast recovery from failures
 Migration	 Move snapshot to new cluster	 Recreate same state in another environment	 Easy cluster migration
 Scheduled Backup	 Automate snapshots (cron job)	 Regular protection of cluster data	 Prevent data loss

Velero Workflow

Cluster Resources  → Velero **Backup** → **Store** (S3)  → **Restore** 

etcd Backup vs Velero

 Feature	 etcd Backup	 Velero
 Scope	 Cluster metadata etcd stores cluster state (Infra)	 Resources +  PVC data Velero = backs up apps + persistent data
 Use Case	 Disaster recovery etcd restores entire cluster	 Workload Backup & migration Velero restores apps selectively
 EKS	 Not needed (AWS Managed clusters, etcd is handled by AWS → no etcd access)	 Yes use Velero for workloads in EKS
 Volumes	 Not supported etcd doesn't store persistent data	 Supported Velero backs up volumes

 Feature	 etcd Backup	 Velero
		(EBS, etc.)

-  etcd = Control plane backup
-  Velero = Application backup

etcd Snapshot vs etcdctl get

 Command	 Purpose	 How It Works	 When to Use
 snapshot	Full backup	 Creates a complete backup of etcd database	Disaster recovery, upgrades
 get	Fetch specific key	 Reads value of a specific key from etcd	Debugging, inspection

Can Velero Backup etcd ?

-  No
-  Velero backs up resources (YAML)
-  etcd snapshot backs up full database

Best Practices

-  Take regular backups
-  Automate with cronjobs
-  Test restore process
-  Secure backup files & Protect etcd certificates

Interview Tip

-  For EKS: In EKS, etcd is managed by AWS. We use Velero + S3 for workload backup.

Final Summary

 Concept	 Meaning
 etcd	Kubernetes database (stores cluster state)
 Backup	Critical for disaster recovery (DR)
 etcdctl	Tool to take etcd snapshots

 Concept	 Meaning
 Velero	Backup workloads + persistent data
 Best Practice	Combine etcd + Velero for full protection

etcd backup protects the Kubernetes control plane, while Velero protects applications and data.